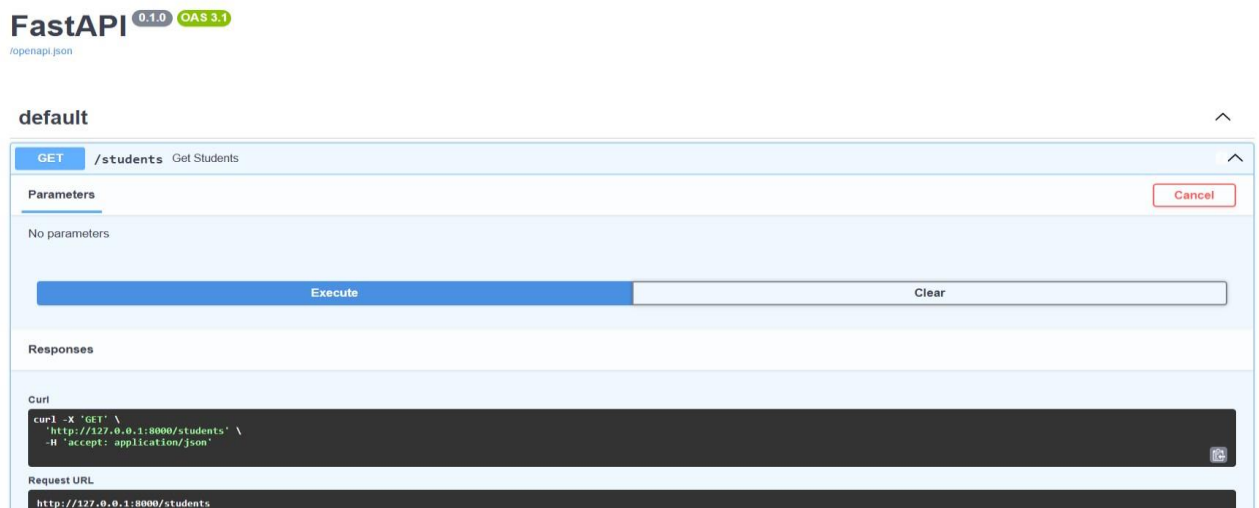


Skill Week-5 (GET & POST Methods)

GET API - Retrieve all students

```
@app.get("/students") def  
get_students():  
    with engine.connect() as connection:  
        result = connection.execute(  
            text("SELECT * FROM students")  
        )  
        students =  
        result.mappings().all()  
  
    return {  
        "message": "Students retrieved successfully",  
        "students": students  
    }
```

What is shown on Swagger for above code-



Server response

Code Details

200

Response body

```
{
  "message": "Students retrieved successfully",
  "students": [
    {
      "id": 0,
      "name": "string",
      "department": "string"
    },
    {
      "id": 101,
      "name": "Ravi",
      "department": "CSE"
    },
    {
      "id": 102,
      "name": "Priya",
      "department": "CSE"
    }
  ]
}
```

Response headers

```
content-length: 194
content-type: application/json
date: Wed, 09 Sep 2026 09:50:04 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

POST API - Add a new student

```
@app.post("/students") def
create_student(student: Student):
    with engine.begin() as
connection:
        connection.execute(
text("""
            INSERT INTO students (id, name, department)
            VALUES (:id, :name, :department)

            """),
        {
            "id": student.id,
            "name": student.name,
            "department": student.department
        }
    )
```

What is shown for above POST code on Swagger-

POST /students Create Student

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "id": 103,
  "name": "Ananya",
  "department": "CSE"
}
```

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/students' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "id": 103,
    "name": "Ananya",
    "department": "CSE"
  }'
```

Request URL

http://127.0.0.1:8000/students

Server response

Code Details

200

Response body

```
{
  "message": "Student created successfully",
  "student": {
    "id": 103,
    "name": "Ananya",
    "department": "CSE"
  }
}
```

Response headers

```
content-length: 98
content-type: application/json
date: Wed, 09 Sep 2026 09:52:16 GMT
server: uvicorn
```

Before Posting Student Data:

Pretty print ☒

```
{
  "message": "Students retrieved successfully",
  "students": [
    {
      "id": 0,
      "name": "string",
      "department": "string"
    },
    {
      "id": 101,
      "name": "Ravi",
      "department": "CSE"
    },
    {
      "id": 102,
      "name": "Priya",
      "department": "CSE"
    },
    {
      "id": 103,
      "name": "Ananya",
      "department": "CSE"
    }
  ]
}
```

After posting student-104:

Pretty print ☒

```
{
  "message": "Students retrieved successfully",
  "students": [
    {
      "id": 0,
      "name": "string",
      "department": "string"
    },
    {
      "id": 101,
      "name": "Ravi",
      "department": "CSE"
    },
    {
      "id": 102,
      "name": "Priya",
      "department": "CSE"
    },
    {
      "id": 103,
      "name": "Ananya",
      "department": "CSE"
    },
    {
      "id": 104,
      "name": "Lavanya",
      "department": "CS&IT"
    }
  ]
}
```

PUT - Update an existing student

```
@app.put("/students/{student_id}") def
update_student(student_id: int, student: Student):
    with engine.begin() as
connection:
        result =
connection.execute(
text("""
            UPDATE students
SET name = :name,
        department = :department
WHERE id = :student_id
        """),
        {
            "student_id": student_id,
            "name": student.name,
            "department": student.department
        }
    )    if
result.rowcount == 0:
    return {
        "message": "Student not found"
    }
    return {
        "message": "Student updated successfully",
        "student": student
    }
```

The screenshot shows a REST client window titled "PUT /students/{student_id} Update Student". It has "Cancel" and "Reset" buttons. The "Parameters" tab is active, showing a table with "Name" and "Description". The "student_id" parameter is marked as "required" and has a value of "104". Below the parameters, the "Request body" is marked as "required" and has a dropdown menu set to "application/json". The "Edit Value" tab is selected, showing a JSON schema for the request body:

```
{
  "id": 104,
  "name": "Priya",
  "department": "CSE"
}
```

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/students/104' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": 104,
    "name": "Priya",
    "department": "CSE"
  }'
```

Request URL

```
http://127.0.0.1:8000/students/104
```

Output after editing using PUT:

```
{
  "id": 104,
  "name": "Priya",
  "department": "CSE"
}
```

DELETE - Delete a student

```
@app.delete("/students/{student_id}") def
delete_student(student_id: int):
    with engine.begin() as
connection:
    result =
connection.execute(
text("""
        DELETE FROM students
        WHERE id = :student_id
        """),
    {
        "student_id": student_id
    }
    )    if
result.rowcount == 0:
    return {
        "message": "Student not found"
    }
```

Output:

DELETE /students/{student_id} Delete Student

Parameters

Name	Description
student_id * required	
integer (path)	103

Execute

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/students/103' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/students/103
```

Server response

Code	Details
200	Response body <pre>{ "message": "Student deleted successfully" }</pre>

Pretty print ☒

```
{
  "message": "Students retrieved successfully",
  "students": [
    {
      "id": 0,
      "name": "string",
      "department": "string"
    },
    {
      "id": 101,
      "name": "Ravi",
      "department": "CSE"
    },
    {
      "id": 102,
      "name": "Priya",
      "department": "CSE"
    },
    {
      "id": 104,
      "name": "Priya",
      "department": "CSE"
    }
  ]
}
```

SQL Workbench Code:

CREATE DATABASE university;

USE university;

CREATE TABLE students (id

INT PRIMARY KEY, name

VARCHAR(100),

department VARCHAR(100)

);

SHOW DATABASES;

USE university;

INSERT INTO students (id, name, department)

VALUES

(101, 'Ravi', 'CSE'),

(102, 'Priya', 'CSE');

SHOW TABLES;

SELECT * FROM students;

Workbench Output:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	id	name	department
▶	101	Ravi	CSE
	102	Priya	CSE
★	NULL	NULL	NULL

students 3 x Apply Revert

Output

Action Output

#	Time	Action	Message
✓ 4	15:17:05	CREATE TABLE students (id INT PRIMARY KEY, name VARCHAR(100), ...	0 row(s) affected
✓ 5	15:17:12	SHOW DATABASES	15 row(s) returned
✓ 6	15:17:33	USE university	0 row(s) affected
✓ 7	15:17:33	SHOW TABLES	1 row(s) returned
✓ 8	15:17:46	INSERT INTO students (id, name, department) VALUES (101, 'Ravi', 'CSE'), (102, '...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
✓ 9	15:17:53	SELECT * FROM students LIMIT 0, 1000	2 row(s) returned